

Samet Tenekeci

Machine Learning Scientist | NLP & Healthcare AI

Izmir Institute of Technology, Turkey

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Summary

Applied AI researcher and engineer with experience developing NLP and machine learning systems for healthcare, biomedical informatics, and software analytics. Delivered multiple AI-driven tools from research to usable software products, including biomedical prediction systems, domain-adapted language models, and data-driven decision support applications. Experienced in leading interdisciplinary teams, managing AI projects, and working with complex real-world datasets in clinical and research environments.

Education

PhD, Computer Engineering 2019 – 2025

Izmir Institute of Technology

Thesis: [Modeling Viral Evolution with Natural Language Processing](#)

MSc, Computer Engineering 2016 – 2019

Dokuz Eylül University

Thesis: [Discovering Disease-Causing Genes by Network Analysis](#)

BSc, Computer Engineering 2009 – 2014

Izmir University

Professional Experience

Researcher & Instructor 06/2025 – Present

Izmir Institute of Technology

- Taught courses in AI, software engineering, and computer science
- Mentored students in interdisciplinary AI research and development projects
- Led projects in Software Engineering & AI Research Group

Research Assistant 07/2018 – 06/2025

Izmir Institute of Technology

- Supported the delivery of computer science and AI courses
- Contributed to several projects in Data Analytics Research Group
- Led administrative tasks within the department as the RA coordinator
- Contributed to the organization of academic conferences and workshops

Machine Learning Engineer 02/2017 – 05/2018

Airties Wireless Networks

- Developed ML-based monitoring and anomaly detection systems for networking devices
- Built data-driven workflows for large-scale overheating detection and mitigation

Teaching Assistant 11/2014 – 08/2016

Izmir University

- Supported the delivery of computer science courses
- Conducted laboratory sessions for undergraduate courses
- Coordinated departmental academic and administrative operations

Selected Projects

MorphoMark

10/2025 – Present

- Developed a deep learning system for automated 3D facial landmark localization using mesh and point cloud data for orthodontic and craniofacial analysis. Built multi-view learning pipelines for anatomical landmark detection and coordinated interdisciplinary AI development efforts involving dentistry and computer engineering teams.

CoV-SNN

06/2023 – 06/2025

- Developed a protein language modeling framework for viral escape prediction and zero-shot variant classification using contrastive learning. Led model development, experimentation, and software implementation for efficient viral sequence analysis and predictive modeling.

AI-Estimator

01/2023 – 01/2025

- Developed domain-adapted language models for automated software analytics and effort estimation. Built NLP pipelines for software project analysis using real-world engineering datasets and contributed to AI-driven analytics tools and predictive modeling workflows.

Selected Publications

1. **Tenekeci S**, Sezgin E, Tekir S. [A contrastive learning framework for efficient viral escape prediction](#). *IEEE Transactions on Computational Biology and Bioinformatics*, 2026.
2. **Tenekeci S**, Tekir S. [Identifying promoter and enhancer sequences by graph convolutional networks](#). *Computational Biology and Chemistry*, 2024.
3. **Tenekeci S**, Işık Z. [Integrative biological network analysis to identify shared genes in metabolic disorders](#). *IEEE Transactions on Computational Biology and Bioinformatics*, 2020.
4. **Tenekeci S**, Ünlü H, Gül BA, Keleş D, Küçük M, Demirörs O. [Automating software size measurement from Python code using language models](#). *Automated Software Engineering*, 2025.
5. Ünlü H, **Tenekeci S**, Kennouche DE, Demirörs O. [Automating software size measurement with language models: Insights from industrial case studies](#). *Journal of Systems and Software*, 2026.

Skills

Domains:

Machine Learning, Deep Learning, Natural Language Processing, Healthcare AI, Biomedical NLP, Clinical Data Analysis, Computational Biology, Multi-omics Data Integration, Software Analytics, High-Performance Computing

Technical:

Python, SQL, R, PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, Hugging Face, Docker, GCP, Linux, Git, SLURM, REST APIs, Streamlit

Service and Community

- Organization Committee Member, IWSM Mensura Conference
- Contributor to national and international grant proposals including Horizon Europe
- Reviewer for Journal of Supercomputing and Knowledge and Information Systems
- Campus Advisor, GitHub
- Member, Data Analytics Research Group
- Member, Software Engineering & AI Research Group

References

- **Assoc. Prof. Dr. Selma Tekir**
PhD Advisor
Department of Computer Engineering, Izmir Institute of Technology
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- **Prof. Dr. Efe Sezgin**
PhD Co-advisor
Department of Molecular Biology and Genetics, Izmir Institute of Technology
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- **Assoc. Prof. Dr. Ekinan Ufuktepe**
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